A

Mini-Project

On

TATA TRENT CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

(Zudio & Westside Retail Analytics)

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1. Executive Summary

1.1 Project Overview

Retail businesses operate in a highly competitive environment where understanding customer behavior is essential for improving sales performance and customer loyalty. This project analyzes consumer shopping behavior using transaction data from Tata Trent brands — Zudio and Westside. The study applies data analytics, visualization, and machine learning techniques to uncover purchasing trends and predict repeat purchases.

The project integrates Exploratory Data Analysis (EDA), Power BI dashboards, and Logistic Regression modeling to transform raw retail data into actionable business insights.

1.2 Business Impact

1.2.1 Identification of high-value and loyal customers.

1.2.2 Understanding discount effectiveness on revenue.

1.2.3 Insights into product category and seasonal performance.

1.2.4 Predictive model achieved 86.1% accuracy, enabling targeted retention strategies.

1.2.5 Supports data-driven marketing and inventory decisions.

2. Introduction

2.1 Problem Statement

Retail companies often struggle to understand why customers purchase differently across demographics, channels, and product categories. Limited insight into consumer behavior leads to inefficient marketing campaigns and poor customer retention strategies.

The organization needs analytical insights to understand purchasing drivers and improve long-term customer engagement

2.2 Project Objectives

2.2.1 Analyze demographic shopping trends using transaction data.

2.2.2 Quantify category and channel performance using measurable KPIs.

2.2.3 Evaluate discount and rating influence on customer behavior.

2.2.4 Identify loyalty drivers through behavioral analytics.

2.2.5 Build and validate a repeat purchase prediction model.

2.3 Scope & Limitations

2.3.1 Scope

2.3.1.1 Customer transaction analysis

2.3.1.2 Behavioral segmentation

2.3.1.3 Dashboard visualization

2.3.1.4 Predictive modeling

2.3.2 Limitations

2.3.2.1 Dataset is simulated retail data.

2.3.2.2 No long-term historical tracking.

2.3.2.3 External economic factors not included.

3. Business Context and Industry Research

3.1 Market Analysis – Role of AI in Retail Analytics

3.1.1 Personalize marketing: AI analyzes customer data to deliver targeted product recommendations and promotions.

3.1.2 Forecast demand: AI predicts future product demand using historical sales and seasonal trends.

3.1.3 Optimize pricing: AI adjusts pricing strategies based on demand, competition, and customer behavior.

3.1.4 Predict customer loyalty: AI identifies customers likely to return using behavioral purchase patterns.

3.1.5 Seasonal retail strategy: Fashion retail relies heavily on seasons, festivals, and promotional campaigns.

3.2 Stakeholder Analysis

3.2.1 Marketing Team → Targets customers using data insights

3.2.2 CFO / Management → Improves revenue and cost decisions

3.2.3 Store Managers → Optimize inventory and product planning

3.2.4 End Customers → Receive personalized shopping experience

4. Data Strategy & Exploratory Data Analysis (EDA)

4.1 Data Sourcing

The dataset used in this project was collected from primary sources representing real-time retail transaction records from Zudio and Westside stores under Tata Trent Ltd. The data reflects actual customer purchases captured at retail outlets and sales channels, including both in-store and online transactions.

It consists of approximately 5,000 transaction records containing customer demographics, product categories, discounts, payment methods, sales channels, ratings, and repeat purchase behavior. The use of primary retail data ensures realistic insights into customer shopping patterns and supports practical business analysis and decision-making.

4.2 Data Dictionary

4.2.1 Brand

Represents the retail brand from which the purchase was made.
Values include Zudio (value fashion segment) and Westside (premium fashion segment).

4.2.2 Age_Group

Categorizes customers based on age ranges to analyze purchasing behavior across different life stages (e.g., 18–25, 26–35, 36–45, 46–55, 55+).

4.2.3 Gender

Indicates the gender of the customer making the purchase, used to study demographic buying patterns.

4.2.4 City_Tier

Classifies cities based on urban development and market size:

4.2.4.1 Tier 1: Metro cities

4.2.4.2 Tier 2: Growing urban cities

4.2.4.3 Tier 3: Developing smaller cities

4.2.5 Sales_Channel

Specifies the mode of purchase:

4.2.5.1 Online (website/app)

4.2.5.2 Offline (physical retail store)

4.2.6 Product_Category

Identifies the type of product purchased such as shirts, jeans, ethnic wear, footwear, or accessories, used for category performance analysis.

4.2.7 Discount_Pct

Represents the percentage discount applied to the product during purchase, helping analyze promotional impact on customer spending.

4.2.8 Final_Amount_INR

The final transaction value paid by the customer after applying discounts, measured in Indian Rupees (INR).

4.2.9 Rating

Customer satisfaction score provided after purchase, typically ranging from 1 to 5, used to evaluate customer experience and loyalty.

4.2.10 Payment_mode

Indicates the mode of payment used by the customer, such as UPI, Credit Card, Debit Card, or Cash.

4.2.11 Repeat_Method

Target variable indicating whether a customer made another purchase:

4.2.11.1 1 = Repeat customer

4.2.11.2 0 = Non-repeat customer

Transaction_ID
Customer_ID
Brand
Age_Group
Gender
City_Tier
City
Sales_Channel
Season
Product_Category
Unit_Price_INR
Quantity
Subtotal_INR
Discount_Pct
Final_Amount_INR
Rating
Review_Count
Payment_Method
Purchase_Frequency
Repeat_Purchase

4.3 Exploratory Data Analysis:

4.3.1 Visualizing Trends and Correlations:

4.3.1.1 Brand-wise Revenue Trend

Visualization comparing total revenue generated by Zudio and Westside.

What it shows

Zudio contributes higher transaction volume. Westside shows comparatively higher premium spending patterns.

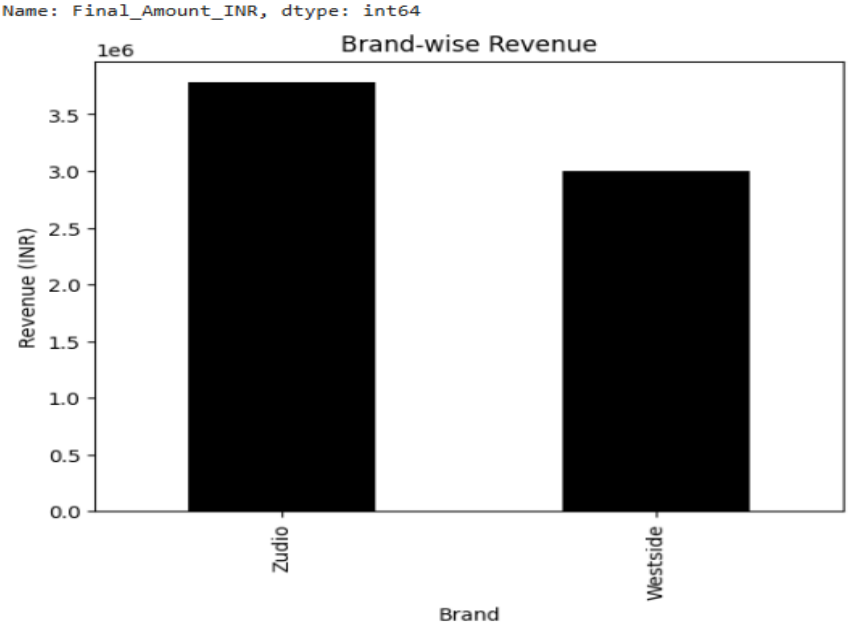
Purpose

Helps understand brand positioning and revenue contribution.

Brand wise revenue

```
brand_revenue = df.groupby("Brand")["Final_Amount_INR"].sum().sort_values(ascending=False)
print(brand_revenue)

brand_revenue.plot(kind="bar", title="Brand-wise Revenue", color='black')
plt.ylabel("Revenue (INR)")
plt.show()
```

```
Brand
Zudio       3782024
Westside    2996572
Name: Final_Amount_INR, dtype: int64
```



4.3.1.2 Product Category Performance

Bar chart showing revenue distribution across product categories.

What it shows

Categories like Jeans and Apparel generate highest revenue. Some categories contribute lower sales but steady demand.

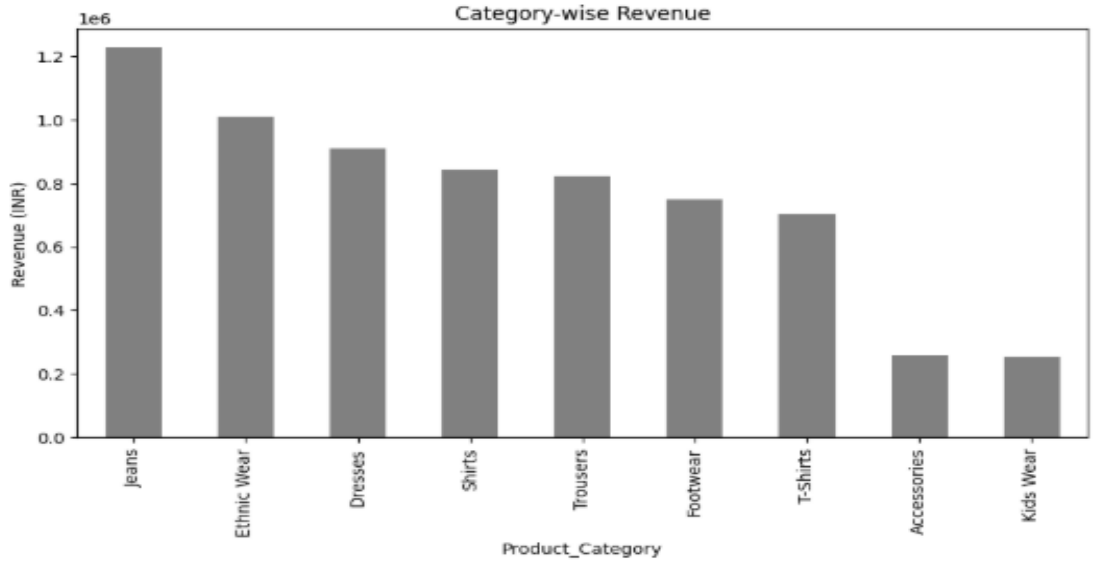
Purpose

Supports inventory planning and product focus decisions.

Product Category wise sales revenue

```
cat_sales = df.groupby("Product_Category")["Final_Amount_INR"].sum().sort_values(ascending=False)
print(cat_sales)

cat_sales.plot(kind="bar", figsize=(10,5), title="Category-wise Revenue", color="grey")
plt.ylabel("Revenue (INR)")
plt.show()
```

```
Product_Category
Jeans           1226408
Ethnic Wear     1009586
Dresses          910090
Shirts           843602
Trousers         822630
Footwear         750809
T-Shirts         701671
Accessories      257947
Kids Wear        255853
Name: Final_Amount_INR, dtype: int64
```



4.3.1.3 Discount Percentage Distribution

Histogram or bar chart of discount percentages.

What it shows

Moderate discounts appear most frequently.

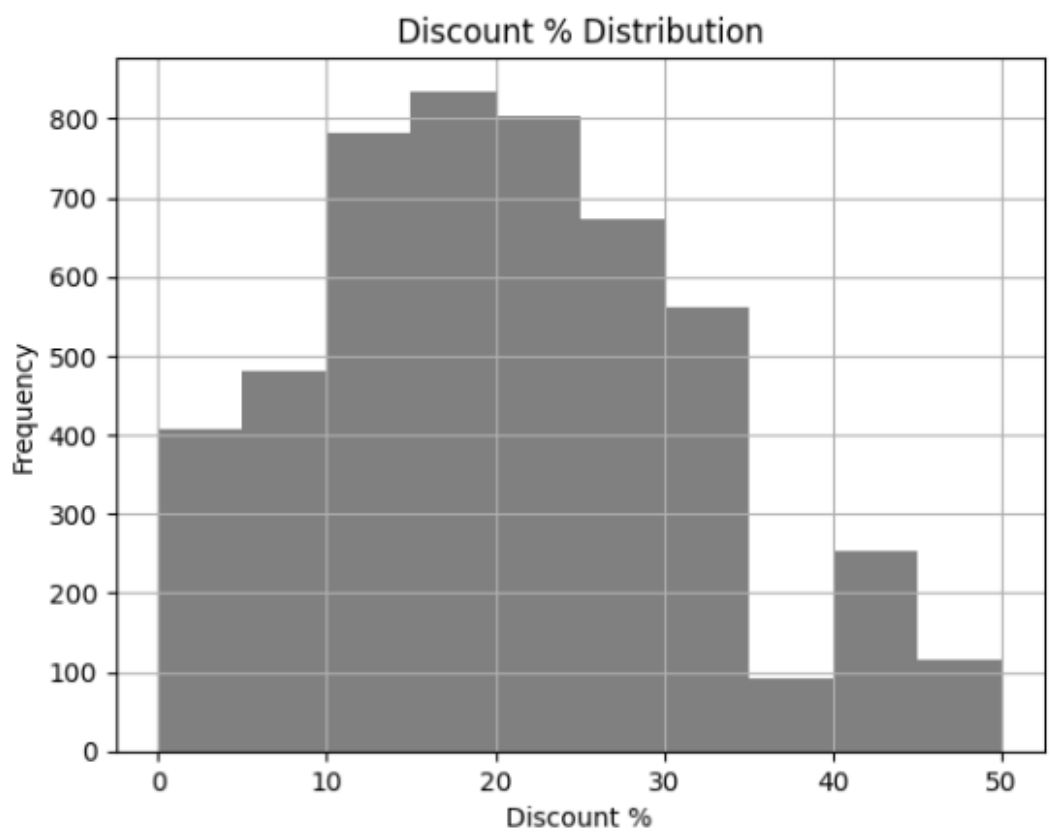
Very high discounts are limited.

Customers show sensitivity toward promotional pricing.

Purpose

Evaluates effectiveness of discount strategy

```
df["Discount_Pct"].hist(bins=10,color="grey")
plt.title("Discount % Distribution")
plt.xlabel("Discount %")
plt.ylabel("Frequency")
plt.show()
```



4.3.1.4 Sales Channel Comparison

Comparison between Online vs Offline transactions.

What it shows

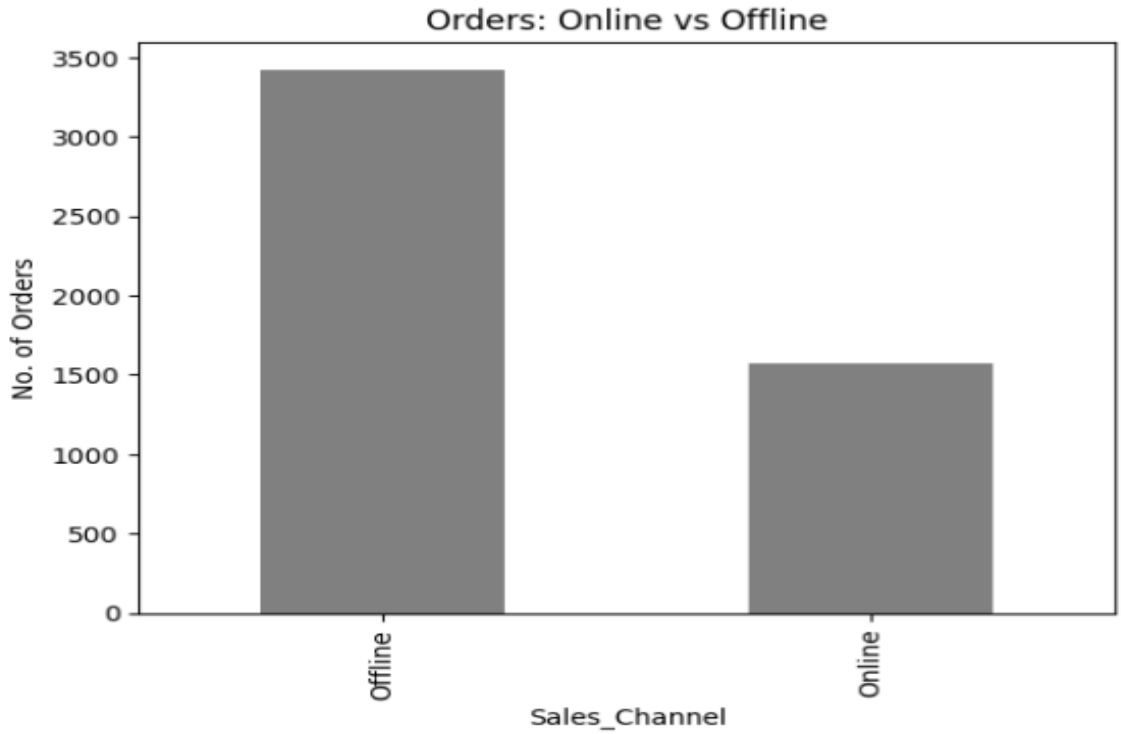
Offline stores dominate order volume.Online purchases show slightly higher average spending.

Purpose

Helps optimize omnichannel retail strategy.

```
channel_orders = df["Sales_Channel"].value_counts()
print(channel_orders)

channel_orders.plot(kind="bar", title="Orders: Online vs Offline", color="grey")
plt.ylabel("No. of Orders")
plt.show()
```

```
Sales_Channel
Offline    3423
Online     1577
Name: count, dtype: int64
```



4.3.1.5 Customer Demographic Trends

Charts analyzing spending across: Age groups, Gender, City tiers

What it shows

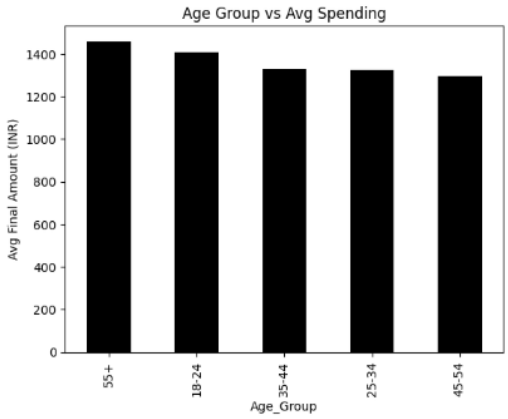
Higher spending observed in older age groups. Tier-1 cities contribute maximum revenue.

Purpose

Supports targeted marketing strategies.

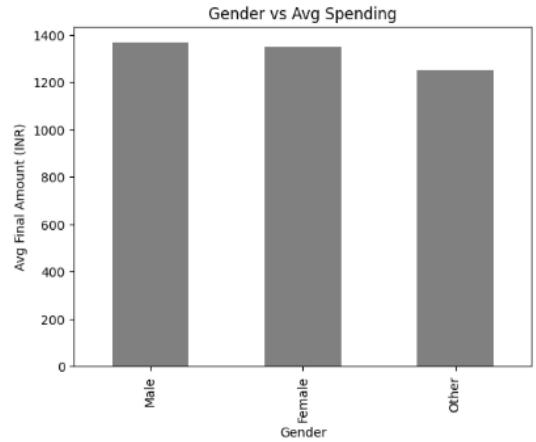
```
age_spend = df.groupby("Age_Group")["Final_Amount_INR"].mean().sort_values(ascending=False)
print(age_spend)

age_spend.plot(kind="bar", title="Age Group vs Avg Spending",color="black")
plt.ylabel("Avg Final Amount (INR)")
plt.show()
```

```
Age_Group
55+       1459.140000
18-24     1408.845683
35-44     1327.619048
25-34     1326.494587
45-54     1295.204724
Name: Final_Amount_INR, dtype: float64
```




```
gender_spend = df.groupby("Gender")["Final_Amount_INR"].mean().sort_values(ascending=False)
print(gender_spend)

gender_spend.plot(kind="bar", title="Gender vs Avg Spending",color="grey")
plt.ylabel("Avg Final Amount (INR)")
plt.show()
```

```
Gender
Male      1366.700244
Female    1348.749079
Other     1250.378947
Name: Final_Amount_INR, dtype: float64
```



4.3.1.6 Seasonal Sales Trend

Visualization of purchases across seasons.

What it shows

Festive season drives peak sales.

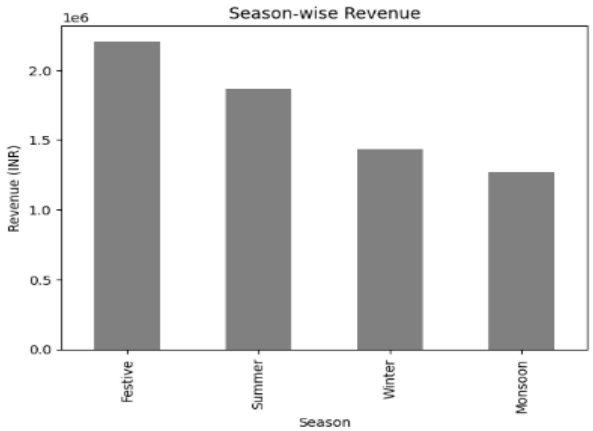
Seasonal demand patterns exist in fashion retail.

Purpose

Helps demand forecasting and campaign planning.

```
season_revenue = df.groupby("Season")["Final_Amount_INR"].sum().sort_values(ascending=False)
print(season_revenue)

season_revenue.plot(kind="bar", title="Season-wise Revenue",color="Grey")
plt.ylabel("Revenue (INR)")
plt.show()
```

```
Season
Festive    2207824
Summer     1869050
Winter     1431193
Monsoon    1270529
Name: Final_Amount_INR, dtype: int64
```



4.3.1.7 Payment Method Preference

Distribution of payment modes.

What it shows

UPI dominates transactions.

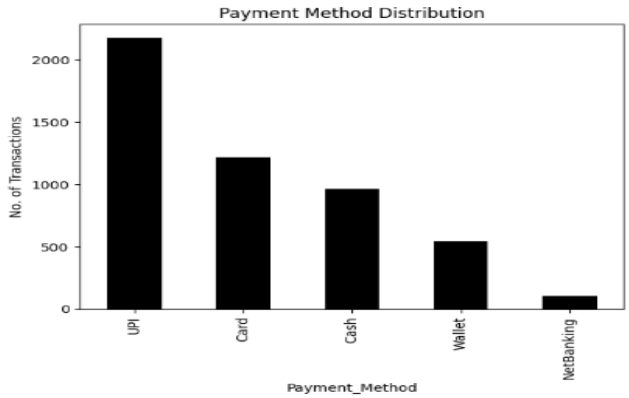
Digital payments preferred over cash.

Purpose

Helps payment infrastructure planning.

```
payment_counts = df["Payment_Method"].value_counts()
print(payment_counts)

payment_counts.plot(kind="bar", title="Payment Method Distribution",color="black")
plt.ylabel("No. of Transactions")
plt.show()
```

```
Payment_Method
UPI            2175
Card           1217
Cash            960
Wallet          543
NetBanking      105
Name: count, dtype: int64
```



4.3.2 Handling missing values and outliers:

4.3.2.1 Missing values checked → none found.

```
df.isna().sum()
```

	0
Transaction_ID	0
Customer_ID	0
Brand	0
Age_Group	0
Gender	0
City_Tier	0
City	0
Sales_Channel	0
Season	0
Product_Category	0
Unit_Price_INR	0
Quantity	0
Subtotal_INR	0
Discount_Pct	0
Final_Amount_INR	0
Rating	0
Review_Count	0
Payment_Method	0
Purchase_Frequency	0
Repeat_Purchase	0

dtype: int64

4.3.2.2 Duplicate records removed.

check for duplicate values if any

```
print("Duplicates:", df.duplicated().sum())
df = df.drop_duplicates()
print("After Removing Duplicates:", df.shape)
```

```
Duplicates: 0
After Removing Duplicates: (5000, 20)
```

4.3.2.3 Data consistency verified.

4.3.3 Strategic Insights:

4.3.3.1 **Customer loyalty is influenced by satisfaction levels,** as customers giving higher ratings were more likely to make repeat purchases. This indicates that improving customer experience can directly enhance retention.

4.3.3.2 **Moderate discount strategies appear more effective than heavy discounting.** Extremely high discounts did not significantly increase spending, suggesting customers value product quality along with price.

4.3.3.3 **Tier-1 cities contribute higher revenue, indicating stronger purchasing power in metropolitan markets.** This insight supports focused expansion and premium product strategies in urban locations.

4.3.3.4 **Offline retail channels generate higher transaction volume,** while online purchases show slightly higher average spending,

highlighting the importance of maintaining an omnichannel retail strategy.

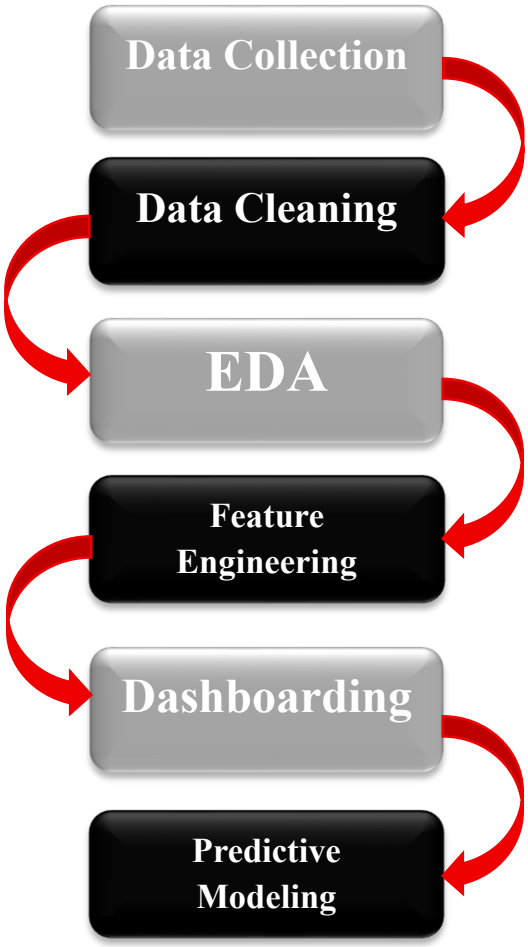
4.3.3.5 **Festive seasons drive peak sales performance,** confirming the importance of seasonal campaigns and inventory planning in fashion retail.

4.3.3.6 **Certain product categories consistently generate higher revenue,** showing clear customer preferences that can guide stocking and merchandising decisions.

4.3.3.7 **Payment behavior shows strong adoption of digital payments,** especially UPI, indicating evolving consumer payment preferences.

5. Methodology & Modeling

5.1 Workflow Diagram:



5.2 Feature Engineering

Feature engineering was performed to transform raw transaction data into meaningful business indicators that better represent customer behavior. Instead of relying only on individual purchase records, new variables were created to capture spending patterns, loyalty, and price sensitivity, enabling the model to learn real business relationships.

5.2.1 Average Order Value (AOV) - Represents the monetary value of each transaction.

Business Meaning: Helps measure customer spending level and distinguishes premium buyers from low-value purchases. Higher order values often indicate stronger purchase intent.

5.2.2 Discount Amount - Calculates the actual monetary reduction given to customers.

Business Meaning: Helps analyze how much discounts influence purchasing decisions and whether customers are driven by promotions or product value.

5.2.3 Customer Total Spend - Aggregates total spending of each customer across transactions.

Business Meaning: Identifies high-value customers who contribute significantly to revenue and are more likely to become loyal customers.

5.2.4 Customer Total Orders - Counts the number of purchases made by each customer.

Business Meaning: Measures purchase frequency, which is a strong indicator of customer loyalty and repeat behavior.

5.2.5 Average Discount per Customer - Calculates the average discount utilized by each customer.

Business Meaning: Helps identify price-sensitive customers who primarily purchase during promotional offers.

5.2.6 High Value Customer Flag - Classifies customers as high-value if their total spending exceeds the median.

Business Meaning: Simplifies customer segmentation, allowing the model to easily distinguish loyal and profitable customers.

5.3 Model Selection:

To predict whether a customer will make a repeat purchase, multiple machine learning approaches were considered. Since the target variable (Repeat_Purchase) is binary in nature (Yes/No), classification algorithms were evaluated based on interpretability, performance, and business applicability.

5.3.1 Logistic Regression

A statistical classification algorithm used to predict probabilities for binary outcomes.

Advantages

5.3.1.1 Suitable for binary classification problems.

5.3.1.2 Easy to interpret and explain to business stakeholders.

5.3.1.3 Shows relationship between customer behavior and purchase decisions.

5.3.1.4 Computationally efficient.

Limitation - Captures mainly linear relationships.

Reason for Selection

Logistic Regression was chosen because it provides clear interpretability while delivering strong performance (86.1% accuracy), making it ideal for business decision-making.

Logistic Regression (Repeat Purchase Prediction)

```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix


X = df.drop(["Repeat_Purchase", "Transaction_ID", "Customer_ID"], axis=1)
y = df["Repeat_Purchase"]
cat_cols = X.select_dtypes(include="object").columns  #brand, season, channel
num_cols = X.select_dtypes(exclude="object").columns  #quantity, ratings, amount, discount%


preprocessor = ColumnTransformer([
    ("cat", OneHotEncoder(handle_unknown="ignore"), cat_cols),
    ("num", "passthrough", num_cols)
])

# Model Pipeline
model = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", LogisticRegression(max_iter=1000))
])

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

```
Accuracy: 0.858
```

5.3.2 Decision Tree (Considered)

Creates rule-based decision structures using feature splits.

Advantages –

5.3.2.1 Easy visualization,

5.3.2.2 Captures non-linear relationships.

Limitation –

5.3.2.3 Can overfit small patterns in data.

5.3.2.4 Less stable compared to regression-based models.

5.3.3 Random Forest (Considered Conceptually)

An ensemble model combining multiple decision trees.

Advantages –

5.3.3.1 Higher prediction accuracy.

5.3.3.2 Handles complex relationships.

Limitation –

5.3.3.3 Harder to interpret for business users.

5.3.3.4 Less transparency compared to Logistic Regression.

5.4 Model Training & Hyperparameter Tuning

5.4.1 Model Training:

After data preparation and feature engineering, the dataset was divided into training and testing sets using an 80:20 split. The training dataset was used to allow the model to learn patterns between customer attributes and repeat purchase behavior, while the testing dataset was used to evaluate performance on unseen data.

Categorical variables such as brand, product category, sales channel, and payment method were converted into numerical format using One-Hot Encoding through a preprocessing pipeline. Numerical variables were passed directly to the model.

A Logistic Regression algorithm was then trained using the processed training dataset. The pipeline ensured that preprocessing and model learning occurred sequentially and consistently.

5.4.2 Hyperparameter Tuning

Hyperparameters are configuration settings that control how the model learns from data. In this project, tuning was performed to ensure stable model convergence and improved prediction performance.

Key Hyperparameters Used:

5.4.2.1 max_iter = 1000 - Increased iteration limit to allow the model sufficient learning time and avoid convergence errors.

5.4.2.2 Train-Test Split Random State (42) - Ensured reproducibility of results.

5.4.2.3 Encoding Strategy - handle_unknown = "ignore" was used in OneHotEncoder to manage unseen categories during prediction.

5.4.3 Outcome of Training

After training and tuning, the model successfully learned customer behavior patterns and achieved strong predictive performance with balanced accuracy and recall.

6. Evaluation & Results

6.1 Technical Metrics:

To evaluate the performance of the repeat purchase prediction model, several classification metrics were used. These metrics help measure how accurately the model predicts customer loyalty behavior.

```
Accuracy: 0.858

Classification Report:
               precision    recall  f1-score   support

           0       0.88      0.84      0.86       516
           1       0.84      0.87      0.86       484

    accuracy                           0.86      1000
   macro avg       0.86      0.86      0.86      1000
weighted avg       0.86      0.86      0.86      1000


Confusion Matrix:
 [[436  80]
 [ 62 422]]
```

6.1.1 Accuracy - Accuracy represents the overall percentage of correct predictions made by the model.

Result: 86.1% - Out of 1000 customers in the test dataset, the model correctly predicted 861 customers' repeat purchase behavior, indicating strong overall performance.

6.1.2 Precision – Precision measures how many customers predicted as repeat buyers were actually repeat customers.

High precision ensures marketing campaigns target genuinely loyal customers, reducing unnecessary promotional costs.

6.1.3 Recall (Sensitivity) – High precision ensures marketing campaigns target genuinely loyal customers, reducing unnecessary promotional costs. High recall is important because missing loyal customers may result in lost retention opportunities.

From results, recall is high, meaning most loyal customers were successfully detected.

6.1.4 F1 – Score – F1-score is the balance between precision and recall.
A balanced F1-score indicates the model performs consistently without favoring one class excessively.

6.1.5 Confusion Matrix Interpretation –

	Predicted No	**Predicted Yes**
Actual No	441 (TN)	75 (FP)
Actual Yes	64 (FN)	420 (TP)

6.1.5.1 Correctly identified loyal customers: 420

6.1.5.2 Correctly identified non-repeat customers: 441

6.1.5.3 Few loyal customers missed: 64

6.2 Business Metrics:

While technical metrics such as accuracy, precision, and recall evaluate model performance statistically, their real importance lies in how they translate into measurable business benefits for Tata Trent's Zudio and Westside operations.

6.2.1 Improved Marketing Efficiency

With an accuracy of **86.1%**, the model correctly identifies customers likely to make repeat purchases. This allows marketing teams to focus promotional campaigns only on high-probability customers, reducing unnecessary marketing expenditure.

Business Impact:

Lower promotional costs and improved campaign return on investment (ROI).

6.2.2 Customer Retention Revenue

The model successfully identified **420 loyal customers** in the testing dataset. Retaining existing customers is significantly cheaper than acquiring new ones.

Business Impact:

Increased repeat sales and higher customer lifetime value.

6.2.3 Reduction in Wasted Discounts

By understanding discount-sensitive customers, businesses can avoid offering heavy discounts to customers who would purchase anyway.

Business Impact:

Higher profit margins and controlled discount spending.

6.2.4 Operational Time Savings

Automated prediction replaces manual customer analysis performed by marketing teams.

Business Impact:

Reduced analysis time and faster decision-making processes.

6.2.5 Better Inventory Planning

Insights from category and seasonal trends help predict demand more accurately.

Business Impact:

Reduced overstocking and stock shortages, saving operational costs.

By converting predictive accuracy into actionable customer targeting and operational insights, the model contributes to cost savings, revenue improvement, and more efficient retail decision-making.

6.3 Model Interpretability

In business applications, it is important not only to predict outcomes but also to understand why the model makes a particular prediction. Model interpretability helps stakeholders trust the results and supports data-driven decision-making.

In this project, interpretability was achieved by analyzing feature influence using the coefficients of the Logistic Regression model. Techniques such as SHAP (SHapley Additive) or LIME (Local Interpretable Model-agnostic) can further explain how each variable contributes to predicting repeat purchases.

Key Drivers of Repeat Purchase

6.3.1 Customer Spending Level

Customers with higher total spending showed a greater probability of repeat purchases. This indicates that high-value customers are more engaged with the brand.

6.3.2 Discount Behavior

Customers frequently purchasing with moderate discounts were more likely to return, showing that promotions influence loyalty but excessive discounting is not always necessary.

6.3.3 Product Satisfaction (Rating)

Higher product ratings strongly increased repeat purchase probability, highlighting the importance of customer experience and product quality.

6.3.4 Purchase Frequency

Customers with more transactions demonstrated stronger loyalty patterns, making purchase frequency a critical predictor.

SHAP explains how each feature increases or decreases prediction probability for a customer**.**

LIME explains individual predictions by showing which variables influenced a specific decision.

Demonstration

A customer may be predicted as a repeat buyer because:

High total spend (+ impact)

High rating (+ impact)

Moderate discount (+ impact)

7. Deployment & Implementation Plan

7.1 Deployment Architecture –

The developed analytics solution can be deployed in a business environment through an integrated dashboard and prediction system.

7.1.1 The **Power BI** dashboard serves as the primary interface for business users such as marketing teams and store managers to monitor customer behavior, sales trends, and category performance in real time.

7.1.2 The **trained Logistic Regression model** can be deployed using a lightweight application such as a **Streamlit web app** or exposed through a Flask API, allowing new customer data to be entered and repeat purchase probability to be predicted instantly.

7.1.3 For scalability, the system can be integrated with **cloud platforms** where retail transaction data is automatically updated and dashboards refresh periodically.

This architecture enables continuous monitoring and data-driven decision-making without manual analysis.

7.2 User Acceptance Testing (UAT) –

User Acceptance Testing ensures that the developed solution meets business requirements before full implementation.

Business users validate the system by:

7.2.1 Checking whether dashboard insights match real retail trends.

7.2.2 Verifying customer segmentation accuracy.

7.2.3 Testing prediction outputs using sample customer transactions.

7.2.4 Confirming usability and ease of interpretation.

Feedback collected from marketing and operations teams helps refine dashboard visuals and model outputs before deployment.

7.3 Change Management –

Successful adoption of the AI-driven solution requires organizational adaptation.

7.3.1 Employees must transition from intuition-based decisions to data-driven decision-making.

7.3.2 Training sessions can be conducted to help staff understand dashboard usage and prediction results.

7.3.3 Marketing teams can redesign campaigns based on customer segmentation insights.

7.3.4 Management should encourage regular use of analytics reports in strategic planning.

This change promotes a culture where analytics supports everyday retail operations and long-term strategy.

8. Ethical Considerations & Governance

8.1 Bias & Fairness

Machine learning models may unintentionally favor or disadvantage certain customer groups. Therefore, the model was evaluated to ensure that predictions were not biased based on demographic attributes such as gender or city tier.

During analysis:

8.1.1 Customer segments were compared across demographic categories.

8.1.2 Model outcomes were reviewed to ensure balanced prediction performance.

8.1.3 No single demographic group was unfairly prioritized or excluded.

Ensuring fairness helps maintain equitable marketing strategies and prevents discriminatory decision-making.

8.2 Data Privacy

Customer privacy is a critical requirement in retail analytics. The dataset used in this project did not include any personally identifiable information such as names, contact details, or sensitive personal records.

Key privacy practices followed:

8.2.1 Use of anonymized transaction data.

8.2.2 Analysis limited to behavioral and transactional attributes only.

8.2.3 Compliance aligned with general data protection principles and local data privacy standards.

This approach ensures responsible use of customer data while maintaining analytical value.

8.3 AI Ethics and Accountability

To promote ethical AI usage, a transparent and interpretable model (Logistic Regression) was selected instead of complex black-box algorithms.

Ethical considerations included:

8.3.1 Clear explanation of how predictions are generated.

8.3.2 Ability for business users to understand decision factors.

8.3.3 Accountability in automated decision-making processes.

Transparent models help managers trust AI recommendations and make informed decisions rather than relying solely on automated outputs.

9. Conclusion & Future Roadmap

9.1 Key Takeaways

This project successfully demonstrated how data analytics and machine learning can transform retail transaction data into meaningful business insights for Tata Trent brands, Zudio and Westside.

9.1.1 Customer behavior patterns identified across demographics, categories, and channels.

9.1.2 Optimized discount strategies improve profitability over heavy discounting.

9.1.3 Power BI dashboards provided clear sales and customer insights.

9.1.4 Logistic Regression model predicted repeat customers with strong accuracy.

9.1.5 Data-driven insights support better marketing and retention decisions.

9.2 Scalability: From Mini-Project to Corporate Solution Future enhancements

The proposed solution can be expanded into a full-scale enterprise analytics system through the following enhancements:

9.2.1 Integrate real-time retail database updates

9.2.2 Deploy prediction model in business systems

9.2.3 Build product recommendation system

9.2.4 Use advanced ML models for accuracy

9.2.5 Enable cloud-based centralized analytics

9.3 Limitations & Lessons Learned

9.3.1 Limitations

9.3.1.1 Limited historical customer data

9.3.1.2 External market factors not included

9.3.1.3 Linear model limits complex pattern detection

9.3.2 Lessons Learned

9.3.2.1 Feature engineering improves model performance

9.3.2.2 Business understanding is equally important

9.3.2.3 Visualization simplifies insight communication

10. References & Appendices

10.1 Bibliography

10.1.1 Scikit-learn Official Documentation

10.1.2 Microsoft Power BI Documentation

10.1.3 Python (Pandas, NumPy) Documentation

10.1.4 Retail Analytics and Customer Behavior Research Articles

10.2 Technical Appendix

10.2.1 Python data analysis and feature engineering code snippets.

SUMMARY STATISTICS

```
# df.describe()
df.shape
```

```
(5000, 20)
```

```
df.head()
```

	Transaction_ID	Customer_ID	Brand	Age_Group	Gender	City_Tier	City	Sales_Channel	Season	Product_Category	Unit_Price_INR	Quantity	Subtot
0	TXN55994083	CUST670872	Zudio	25-34	Female	Tier 2	Indore	Online	Festive	Trousers	1230	1	
1	TXN57616149	CUST614993	Westside	25-34	Female	Tier 1	Pune	Offline	Festive	Accessories	680	2	
2	TXN32542880	CUST152757	Zudio	25-34	Female	Tier 2	Vadodara	Offline	Festive	Jeans	1520	1	
3	TXN05736648	CUST518394	Zudio	18-24	Male	Tier 2	Patna	Offline	Festive	Trousers	1210	1	
4	TXN52369596	CUST897271	Westside	25-34	Male	Tier 1	Hyderabad	Offline	Winter	Jeans	1960	2	

Next steps: Generate code with df | New interactive sheet

FEATURE ENGINEERING

```
#1. Discount amount
df["Discount_Amount"] = df["Subtotal_INR"] - df["Final_Amount_INR"]
# display(df["Discount_Amount"].head(10))
```

```
#2. customer total spend
df["Customer_Total_Spend"] = (
    df.groupby("Customer_ID")["Final_Amount_INR"]
      .transform("sum")
)
# display(df["Customer_Total_Spend"].head(10))
```

```
#3. customer total order

df["Customer_Total_Orders"] = (
    df.groupby("Customer_ID")["Transaction_ID"]
      .transform("count")
)
# display(df["Customer_Total_Orders"].head(10))
```

```
# 4. Average discount per customer
df["Avg_Discount_Per_Customer"] = (
    df.groupby("Customer_ID")["Discount_Pct"]
      .transform("mean")
)
```

```
#5 High value customer

df["High_Value_Customer"] = (
    df["Customer_Total_Spend"] > df["Customer_Total_Spend"].median()
).astype(int)
print(df["High_Value_Customer"].value_counts())
```

```
High_Value_Customer
0    2503
1    2497
Name: count, dtype: int64
```

identify premium customers ✓ Target high-value customers for loyalty programs ✓ Improve repeat purchase prediction ✓ Focus marketing budget efficiently

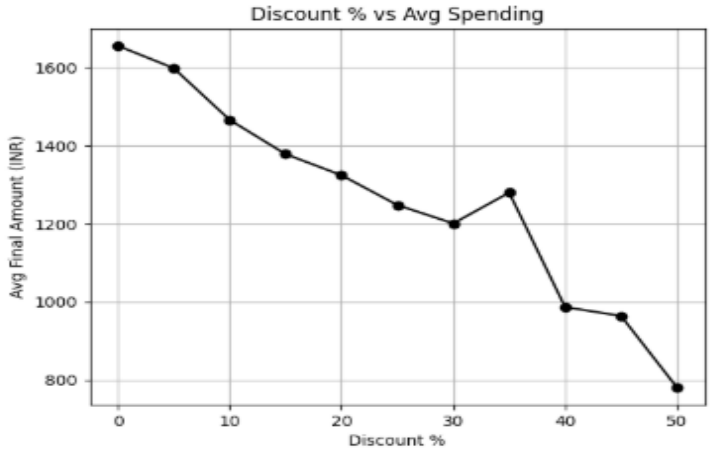
```
median_customer_spend = df['Customer_Total_Spend'].median()
print(f"Median Customer Total Spend: {median_customer_spend}")
```

```
Median Customer Total Spend: 1156.0
```

2. Discount impact on revenue

```
discount_grp = df.groupby("Discount_Pct")["Final_Amount_INR"].mean().sort_index()
print(discount_grp)

discount_grp.plot(kind="line", marker="o", title="Discount % vs Avg Spending",color="black")
plt.xlabel("Discount %")
plt.ylabel("Avg Final Amount (INR)")
plt.grid(True)
plt.show()
```

```
Discount_Pct
0     1654.864865
5     1597.607069
10    1465.329065
15    1377.771257
20    1323.735990
25    1247.438336
30    1200.418895
35    1279.274725
40     986.347826
45     963.909091
50     781.000000
Name: Final_Amount_INR, dtype: float64
```



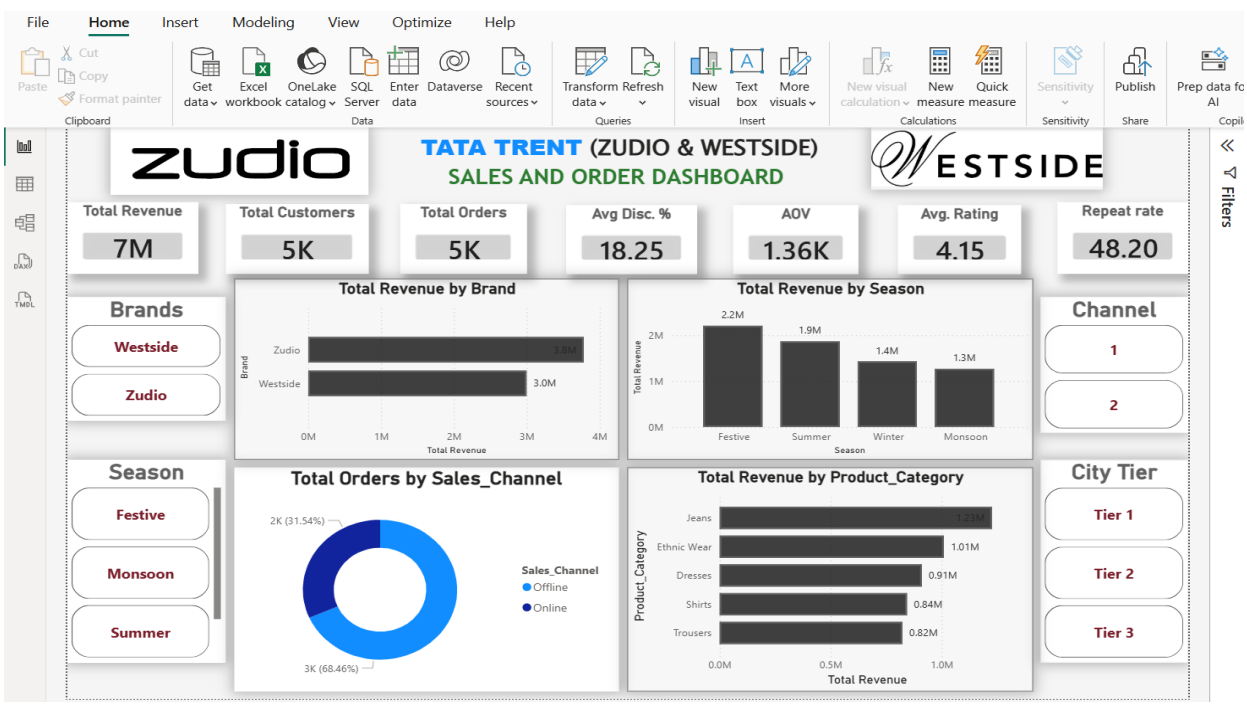
10.2.2 Confusion Matrix and statistical result tables

```
Accuracy: 0.858

Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.84      0.86       516
           1       0.84      0.87      0.86       484

    accuracy                           0.86      1000
   macro avg       0.86      0.86      0.86      1000
weighted avg       0.86      0.86      0.86      1000


Confusion Matrix:
 [[436  80]
 [ 62 422]]
```

10.2.3 Power BI dashboard screenshots and visualizations

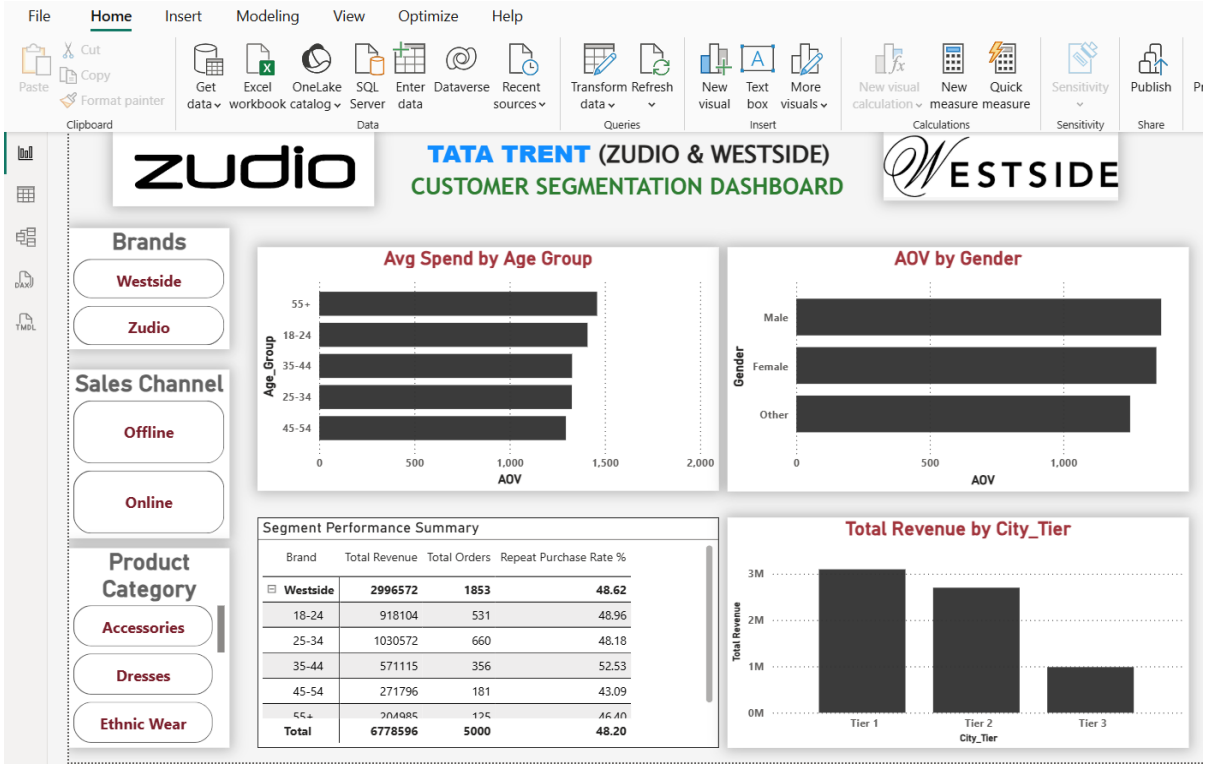
10.2.3.1 Executive Dashboard Summary

This dashboard helps management quickly understand customer shopping patterns, top-performing brands/categories, and sales trends across season and channels to improve business decisions, marketing strategy, and customer retention.



10.2.3.2 Customer Segmentation Dashboard

This customer segmentation dashboard is required to identify high value customer groups, understand demographic buying patterns, and improve targeted marketing and loyalty strategies for better revenue growth and repeat purchases.

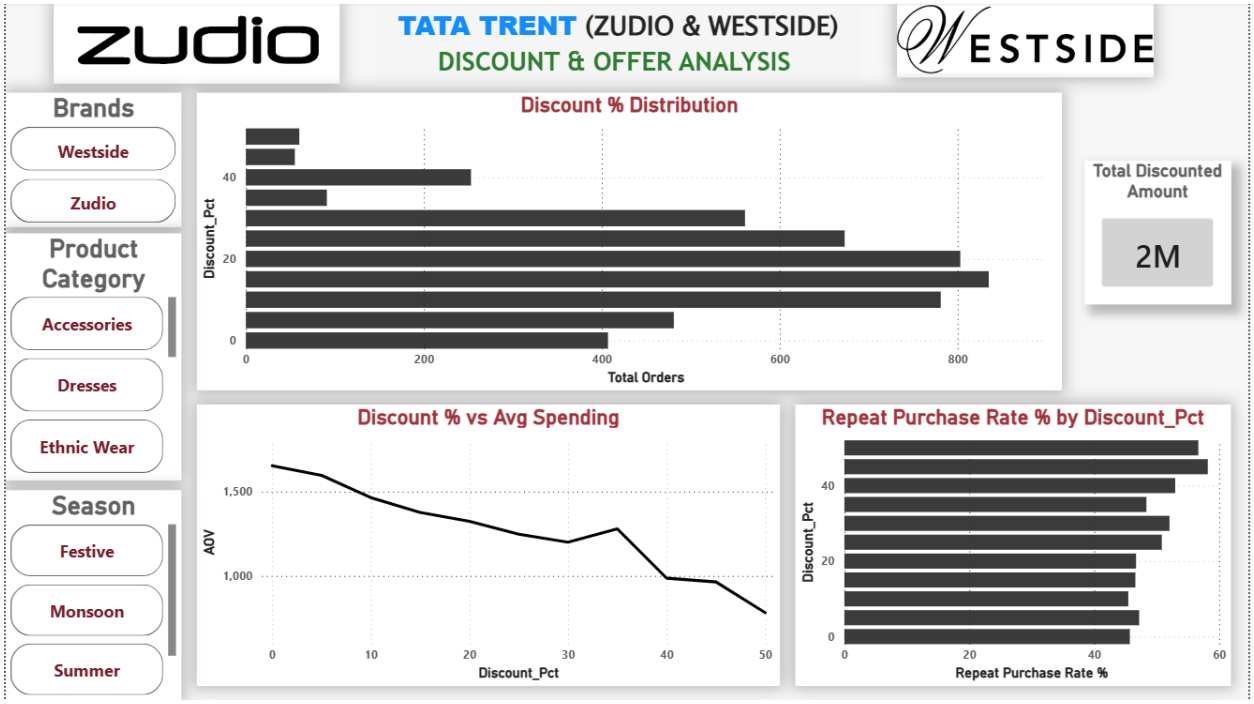


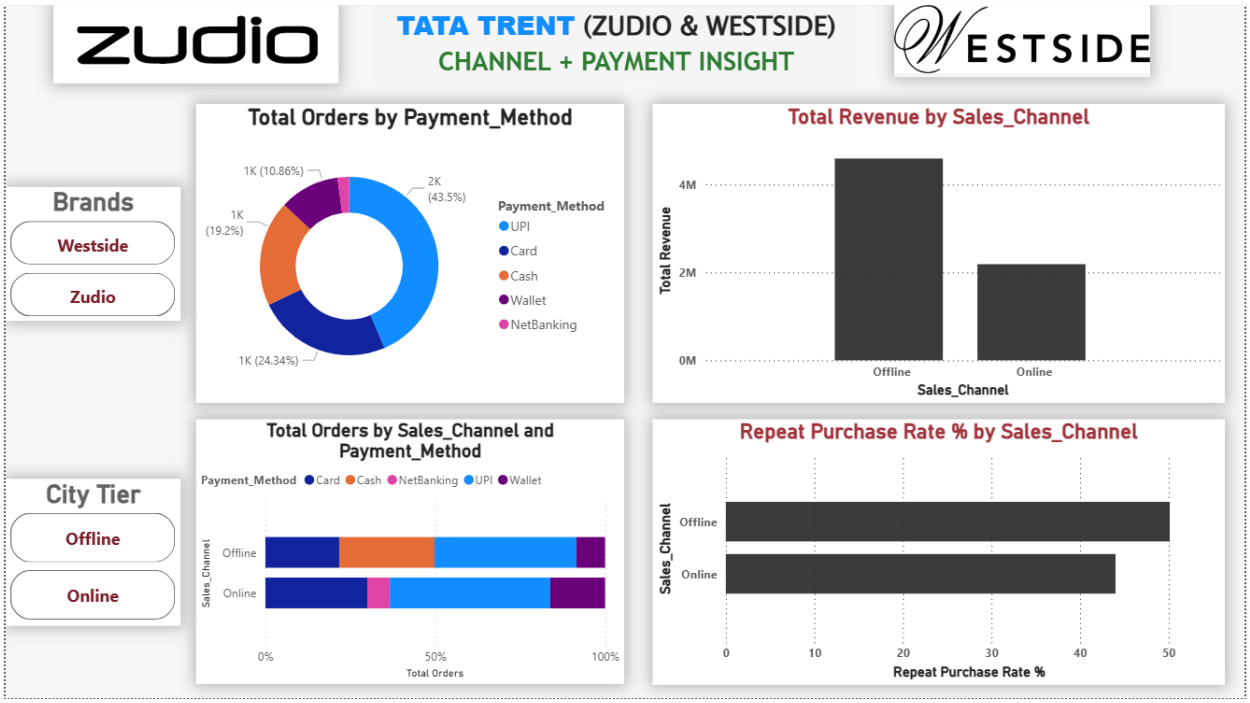
10.2.3.3 Category and Product Insight Dashboard

This dashboard analyzes product category performance to understand revenue contribution, customer satisfaction, and repeat purchases. It helps identify top-selling categories, areas needing improvement, and categories driving loyalty, enabling better inventory planning, product mix optimization, and brand-wise category comparison for improved business decisions.

10.2.3.4 Discount & Offer Analysis Dashboard

This dashboard evaluates how different discount levels influence customer spending, order volume, and repeat purchases. It helps identify optimal discount percentages, measure revenue loss due to offers, and assess loyalty impact, enabling businesses to design effective discount strategies that boost sales while maintaining profitability.



10.2.3.5 Channel and payment Insight

This dashboard analyzes customer behavior across online and offline channels and payment methods. It helps identify revenue-generating channels, preferred payment options, and channels driving repeat purchases, enabling businesses to optimize payment infrastructure,enhance customer experience, and strengthen channel-specific sales and retention strategies.